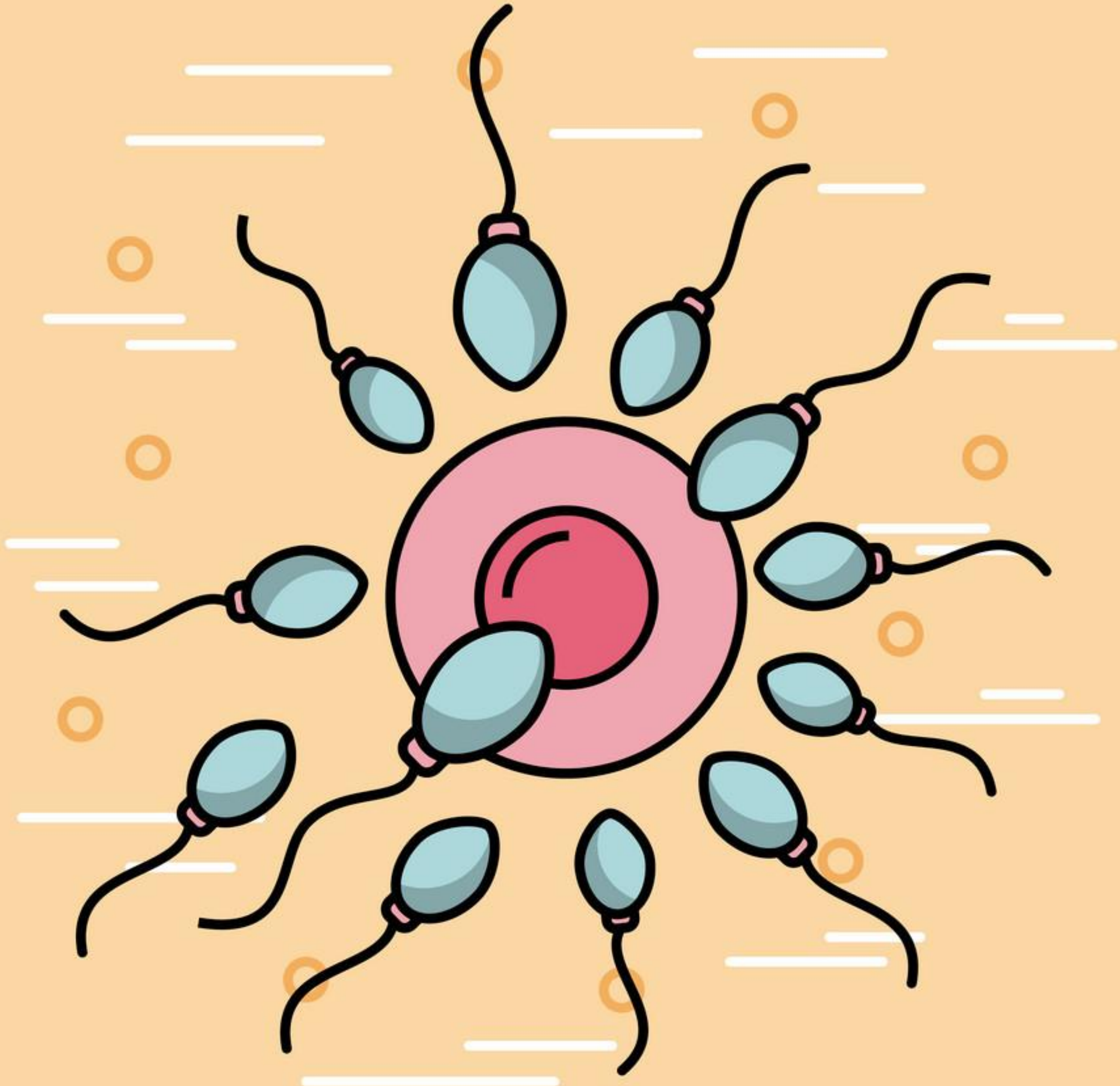


INTRODUCTION TO REPRODUCTIVE PHYSIOLOGY

Dr. Sandra Christina George
Senior Resident

REPRODUCTION

- Process by which new members of a species are produced
- Ensures survival of the species
- **Reproductive health,** according to WHO, is a state of complete, physical, mental and social wellbeing, not merely absence of Reproductive disease or infirmity

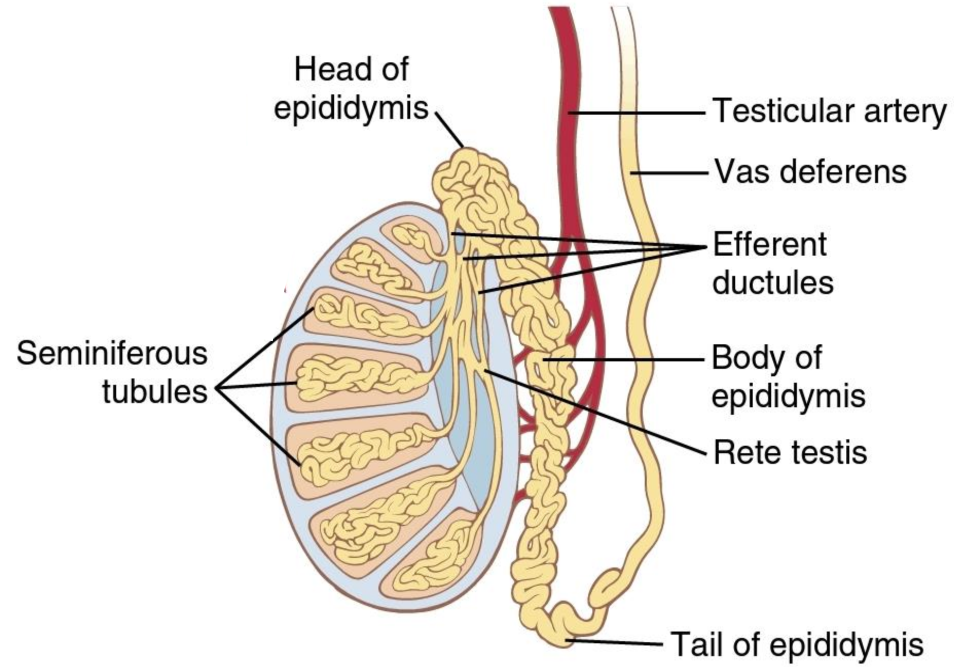
Sex?

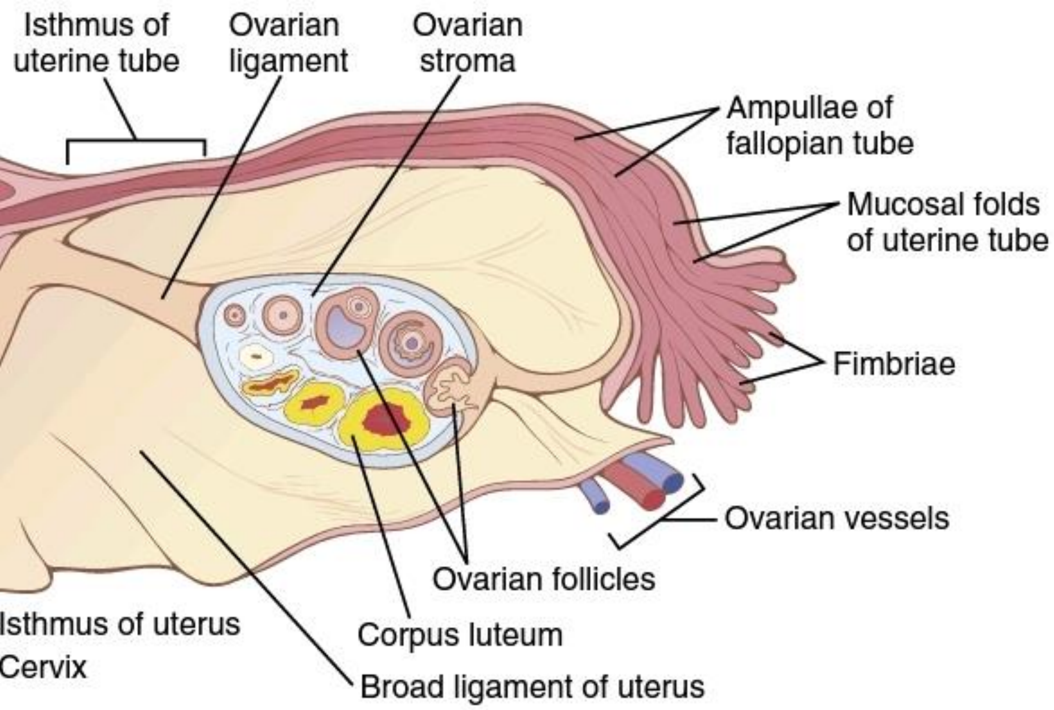
Gender?

Gender Identity?

Sexual Orientation?

Characteristics of sex



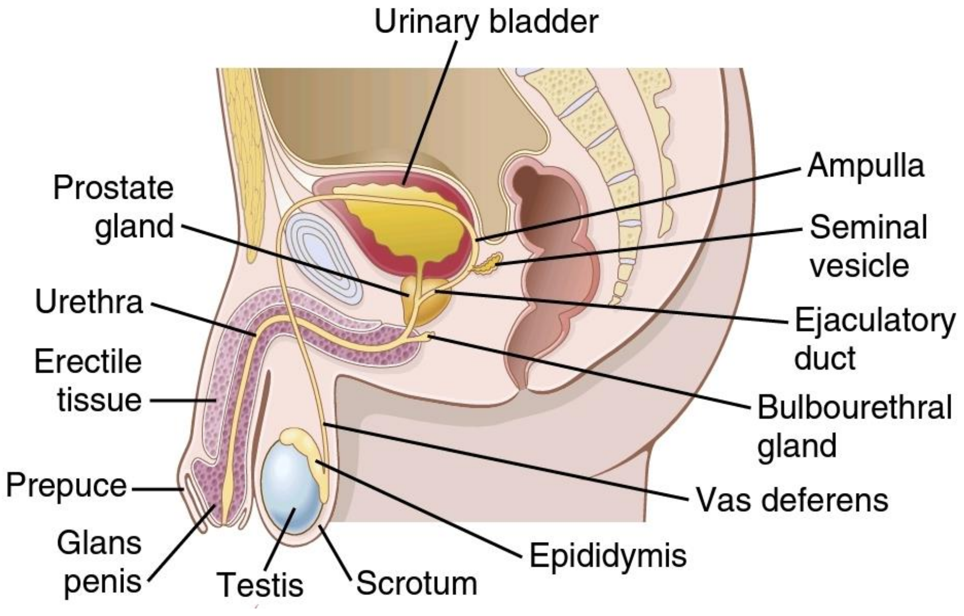


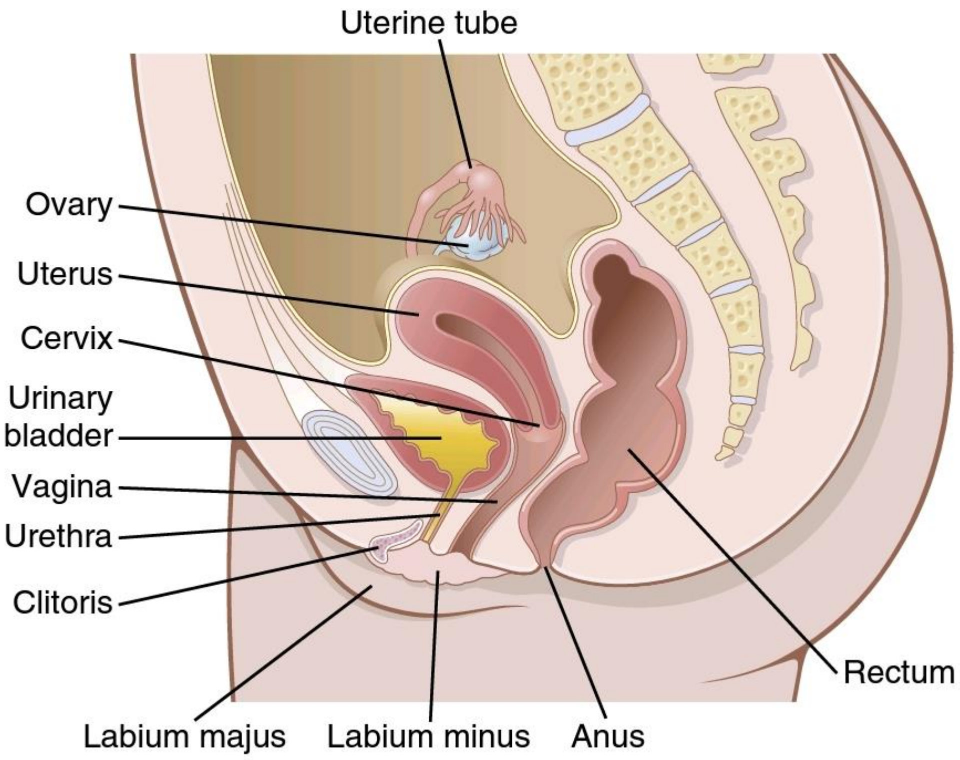
Primary sex organs: GONADS

- Includes a pair of testis in male, a pair of ovaries in females
- Dual function ➔ Gametogenic + secretory (endocrine) functions

Accessory sex organs: INTERNAL and EXTERNAL GENITALIA

- Organs other than gonads, necessary for reproductive function
- Present at birth, enlarge during puberty



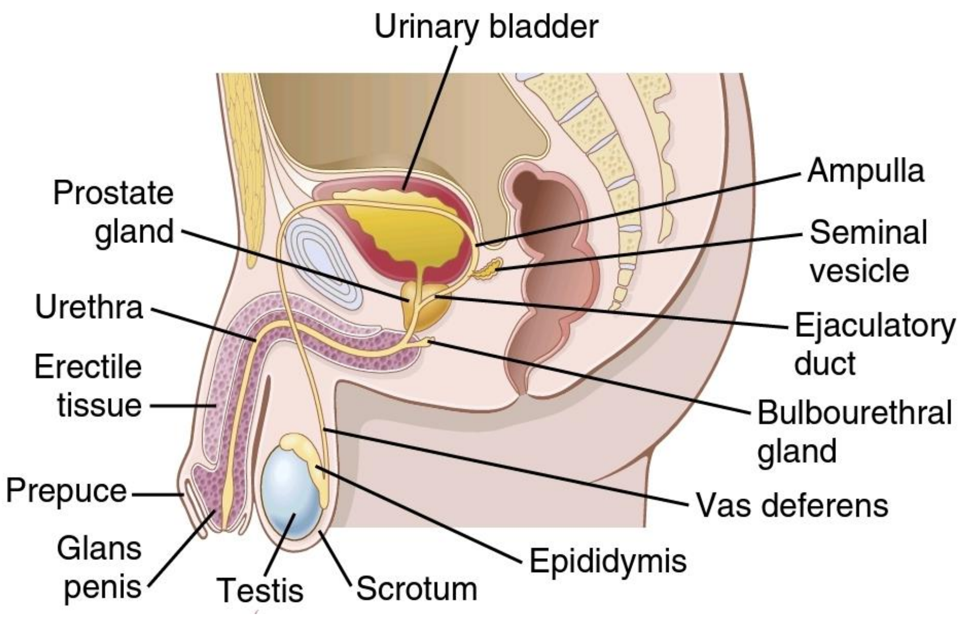


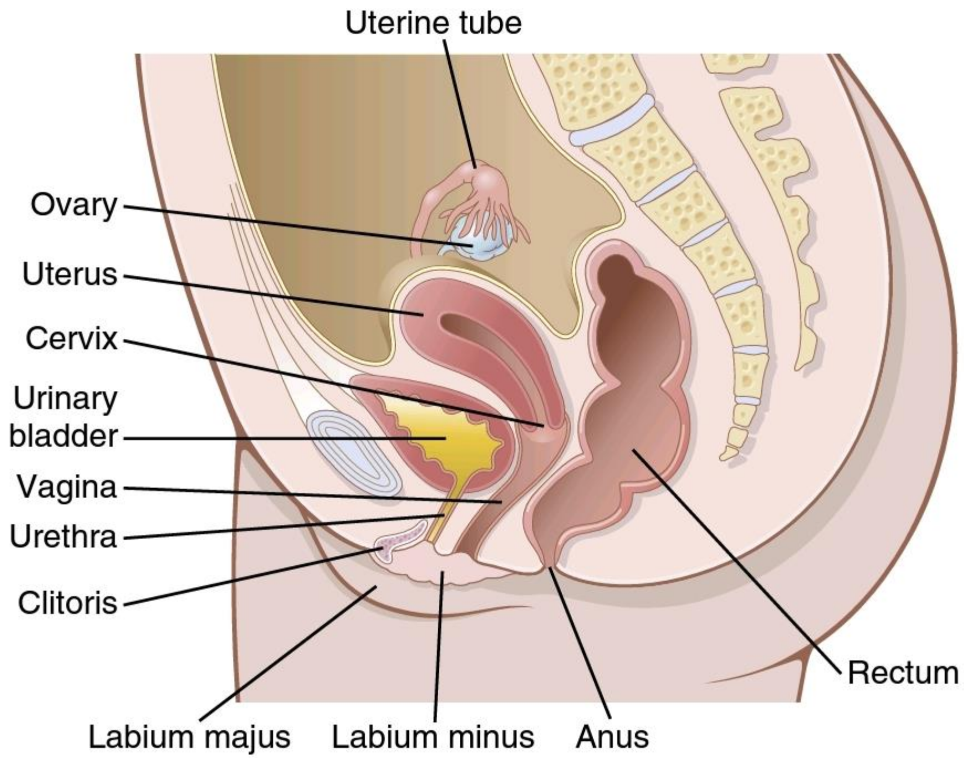
Internal genitalia

- Males → epididymis, vas deferens, seminal vehicle, prostate gland, Cowper's gland
- Females → ovary, uterus, vagina

External genitalia

- Males → urethra, penis, scrotum
- Females → clitoris, labia majora, labia minora, lower vagina





Secondary sexual characteristics

- Differentiate the semester
- Absent at birth and appear at puberty
- Include changes in hair distribution, muscular and mammary gland development, voice, body configuration and mental development

Sexual development

Sex determination

Sex differentiation

Sex determination

Chromosomal sex / genetic differentiation

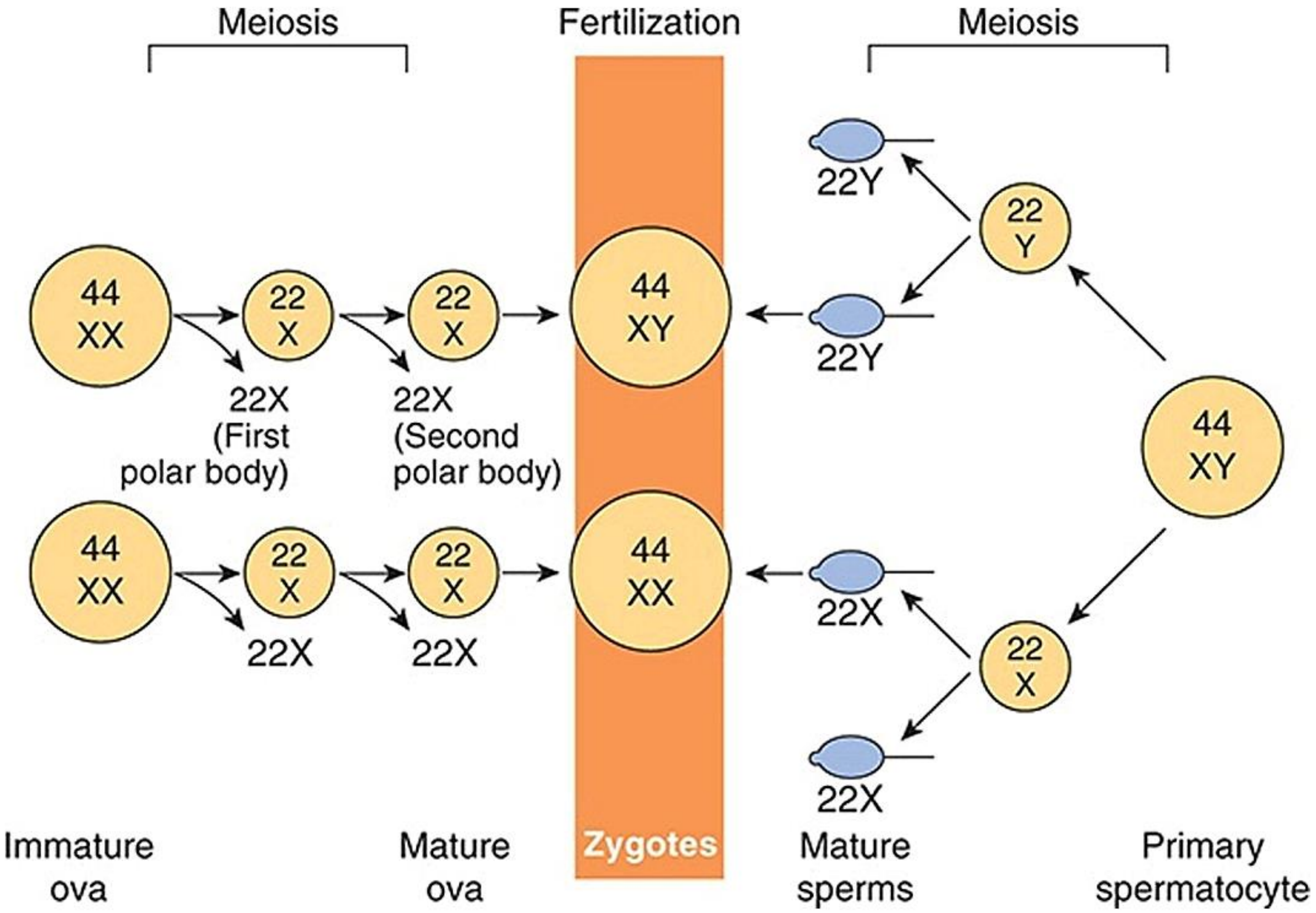
Sex chromosomes → X, Y

Males → 44 autosomes + XY

Females → 44 autosomes + XX

Genetic sex is determined entirely by sperm

Sperm	Ovum	Offspring	Sex genotype
X	X	XX	Genetic female
Y	X	XY	Genetic male


Meiosis
Fertilization
Meiosis
44
XX
22
X
22
X
44
XY
22Y
22
Y
22Y
44
XY
22X
(First
polar body)
22X
(Second
polar body)
44
XX
22
X
22
X
44
XX
22X
22
X
22X
22X
22X
22X
Immature
ova
Mature
ova
Zygotes
Mature
sperms
Primary
spermatocyte

????

No. of Males born > Females born

- Y chromosome is smaller than X chromosome
- Sperms with Y are lighter, swim faster → reaches ovum rapidly

Sex chromatin (Barr Body)

- During embryonic development in females, somatic cells multiply → one X chromosome becomes inactive while other remains active
- Inactive X chromosome condenses to chromatin mass near inner surface of nuclear membrane

- **Cause**: Presence of Xist gene (X inactivation centre) in the chromosome → non coding mRNA → inactivation
- Inactivation is random

- Best seen in squamous epithelial cells of buccal mucosa
- Also seen in 1-15% neutrophils → drumstick appearance

- In adult female, active X chromosome of somatic cells → ½ paternal origin + ½ maternal origin
- No. of Barr bodies = no. of X chromosomes – 1

- **Significance**
  - Identification of sex genotype
  - Identification of abnormal genotype

Sex differentiation

Gonadal differentiation

Genital differentiation

Psychological differentiation

Gonadal differentiation

- Gonadogenesis / gonadal sex
- Formation of gonads: testes in males & ovaries in females

- Primitive gonad → from genital ridge
- Consists of cortex, medulla, primordial germ cells

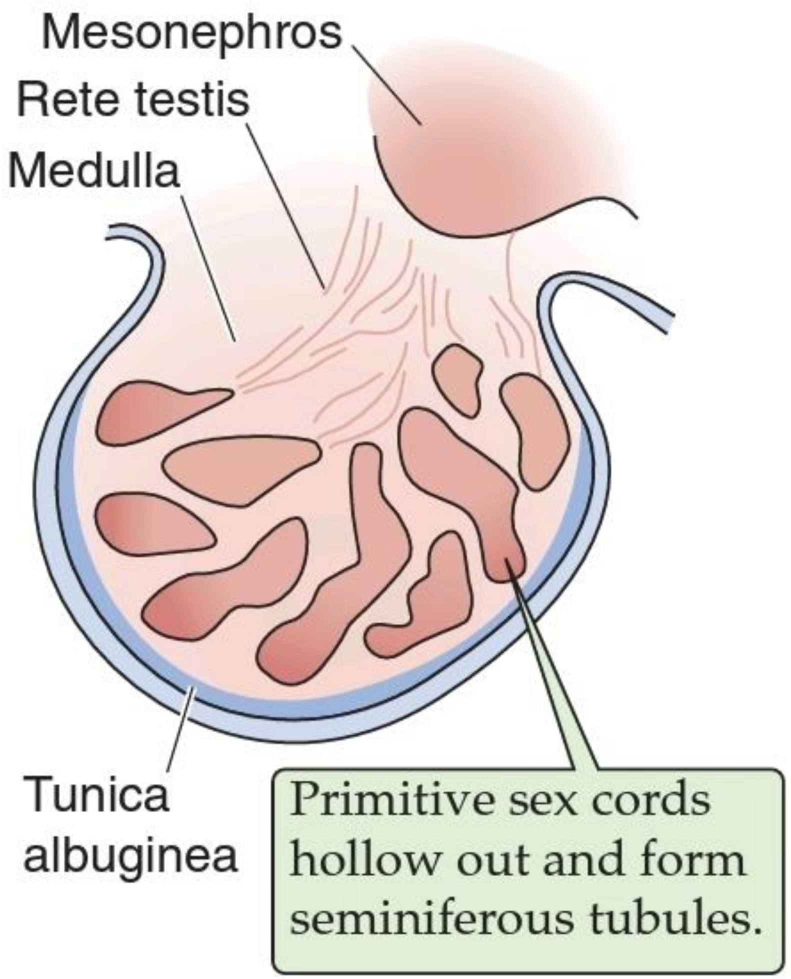
- Bipotential / primordial / primitive / ambisexual gonads
- Upto 6 weeks → identical in both sexes

- After differentiation, testis can be identified by 8th week of IUL and ovary can be identified by 11th / 12th week of IUL

Testicular differentiation

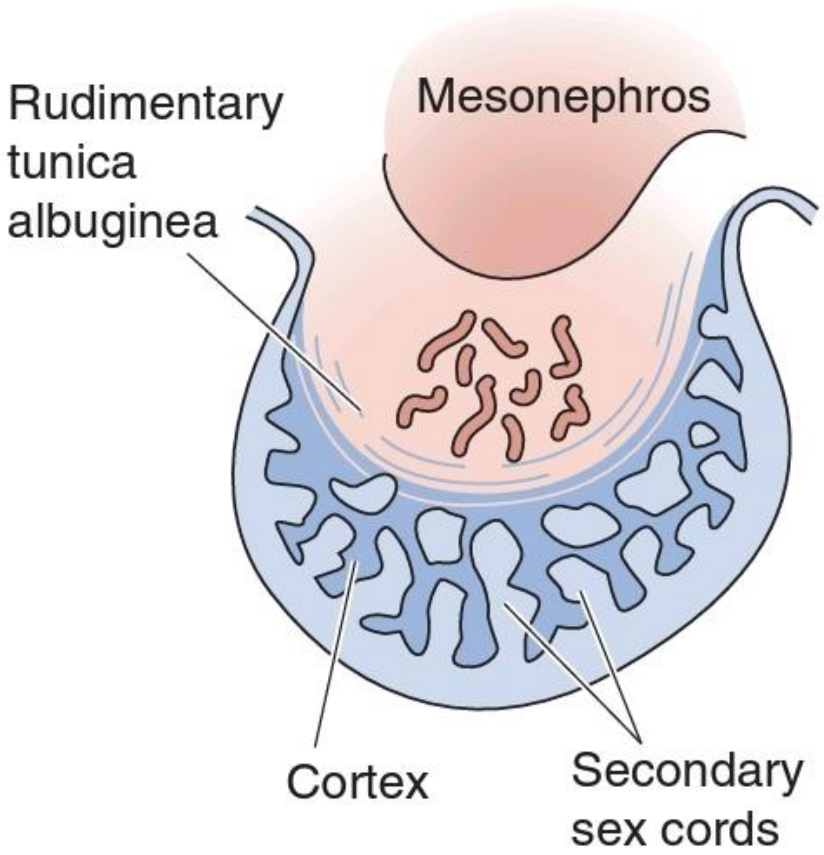
- Y chromosome → important in testicular differentiation
- 2 Transcription genes on Y chromosome
  - SRY gene (Sex determining region of Y chromosome): Testis determining factor (TDF)
  - MIS (Mullerian duct Inhibiting Substance)

6th week	Appearance of sex cords from germinal epithelium
7th week	Sex cords → seminiferous tubule Sertoli cells appear → MIS Regression of cortex
8th week	Leydig cells appear and proliferate
9th week	Leydig cells → testosterone



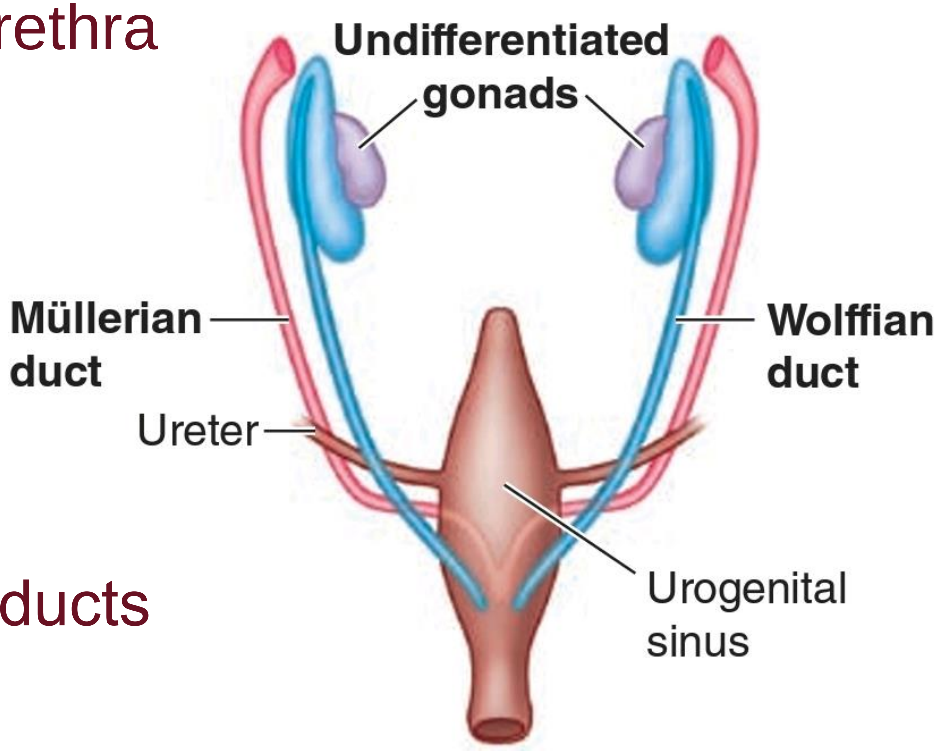
Ovarian differentiation

- Requires presence of X chromosome + absence of TDF
- Ovary develops from cortex, medulla regresses
- Embryonic ovary does not secrete any hormones
- Cortical cells proliferate to form granulosa cells → surround the germ cells
- Germ cells → oogonia → oocyte (11-12 weeks)



Genital differentiation

- Phenotypic sex / genital sex
- Differentiation of internal & external genitalia & urethra
- Dependent upon hormones secreted by the developing gonads



Internal genitalia

7th week of IUL: Embryo → both primordial genital ducts → paired set of Wolffian (male) ducts and Mulleran (female) ducts

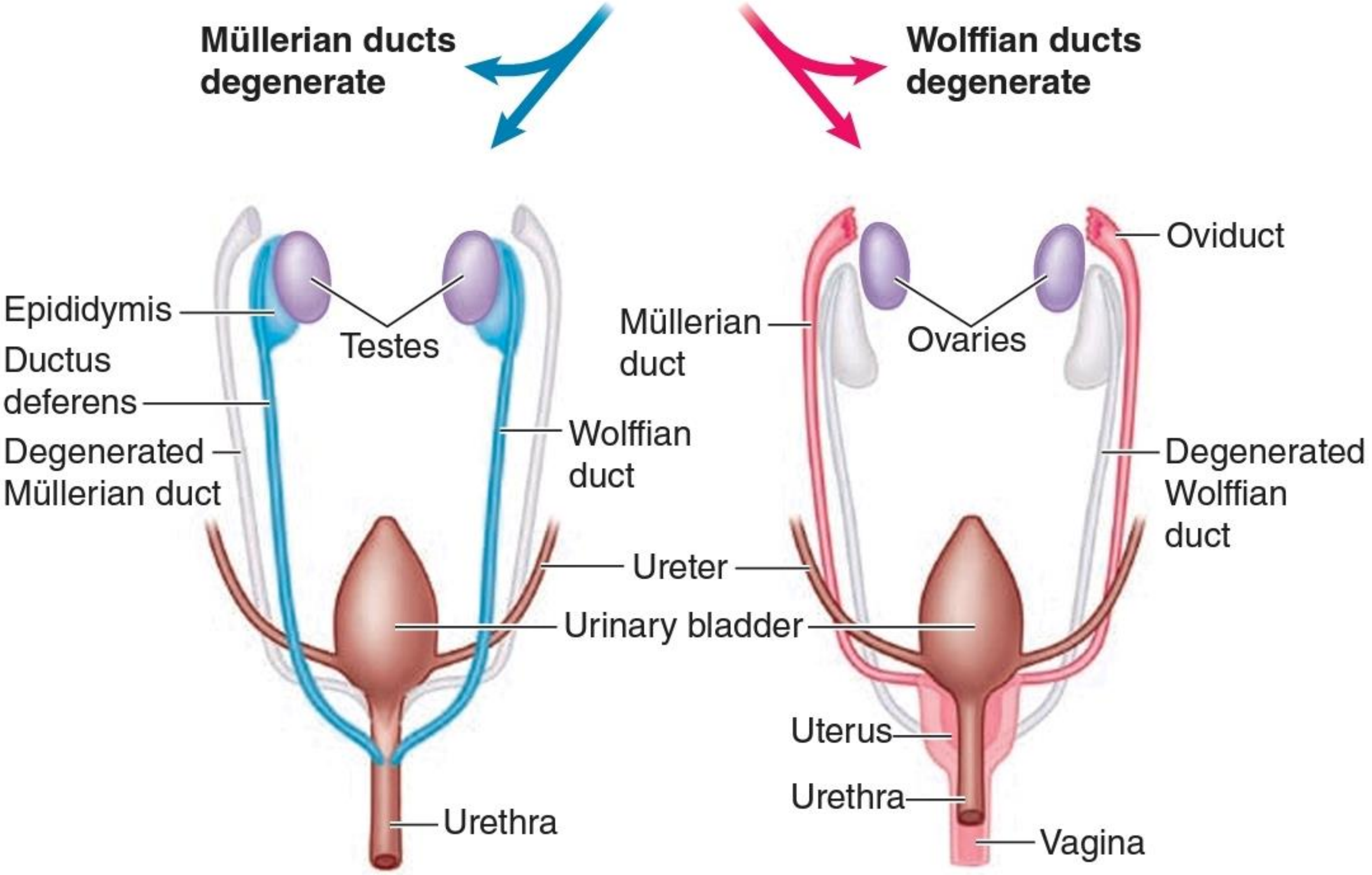
Male internal genitalia

Testis produces MIS and testosterone

- MIS → regression of Mullerian duct by apoptosis
- Testosterone → Wolffian duct proliferates → Epididymis, Vas deferens, Seminal vesicle

Female internal genitalia

- No MIS → Mullerian duct proliferates → Uterine tube, uterus, upper 2/3rd of vagina
- No Testosterone → Wolffian Duct degenerates



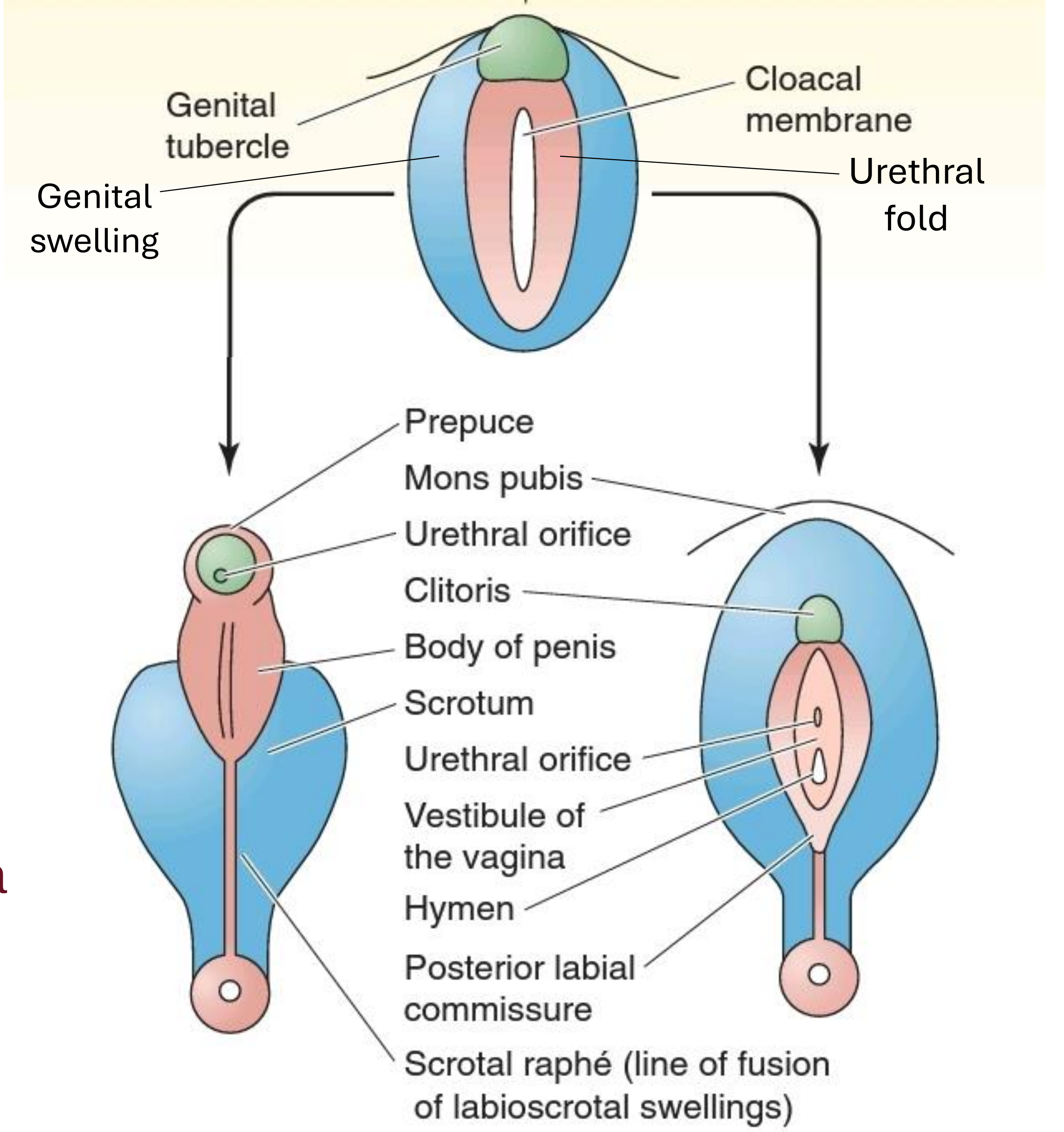
Müllerian ducts degenerate
Wolffian ducts degenerate
Epididymis
Ductus deferens
Degenerated Müllerian duct
Testes
Wolffian duct
Ureter
Urinary bladder
Urethra
Oviduct
Müllerian duct
Ovaries
Degenerated Wolffian duct
Uterus
Urethra
Vagina

External genitalia

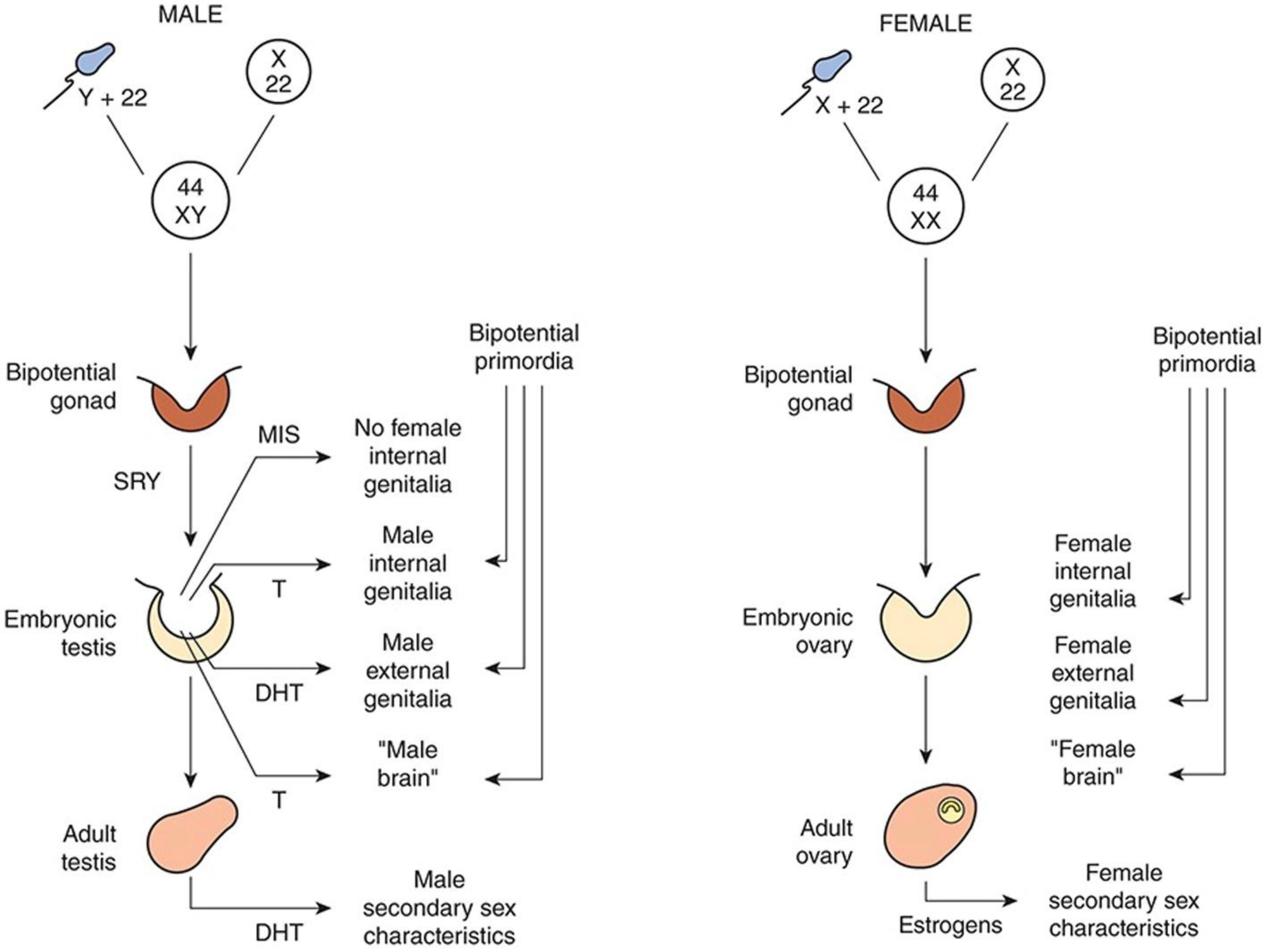
Bipotential until 8th week

Testosterone & Dihydrotestosterone

- If present → Male external genitalia by 5th month of IUL
- If absent → Female external genitalia



	Male derivative	**Female derivative**
Urogenital sinus	Prostate and prostatic urethra	Urethra
Urethral fold	Penile urethra and shaft of penis	Labia minora
Genital swelling	Scrotum	Labia majora
Genital tubercle	Glans penis	Clitoris


MALE
Y + 22
X
22
44
XY
Bipotential gonad
SRY
Embryonic testis
Adult testis
MIS
No female internal genitalia
T
Male internal genitalia
DHT
Male external genitalia
T
"Male brain"
DHT
Male secondary sex characteristics
Bipotential primordia
FEMALE
X + 22
X
22
44
XX
Bipotential gonad
Embryonic ovary
Adult ovary
Estrogens
Female secondary sex characteristics
Bipotential primordia
Female internal genitalia
Female external genitalia
"Female brain"

Psychological differentiation

- Difference in male and female behaviour → anatomical and functional differences in certain areas of brain
- Also determined by effect of androgens on development of brain early in life

- Exposure of fetal HT to androgens → male patterns of sexual behavior
- Absence of androgens in female embryo → female patterns of sexual behavior

Aberrant sexual development

Chromosomal / genetic abnormalities

Developmental / hormonal abnormalities

Chromosomal abnormalities

- Nondisjunction of sex chromosomes during gametogenesis
- Phenomenon in which a pair of chromosomes fail to separate, so that both go to one of the daughter cell during meiosis
- Usually occurs during 1st meiotic division

Trisomy

- 44 XXX
- 44 XXY

Monosomy

- 44 XO
- 44 YO

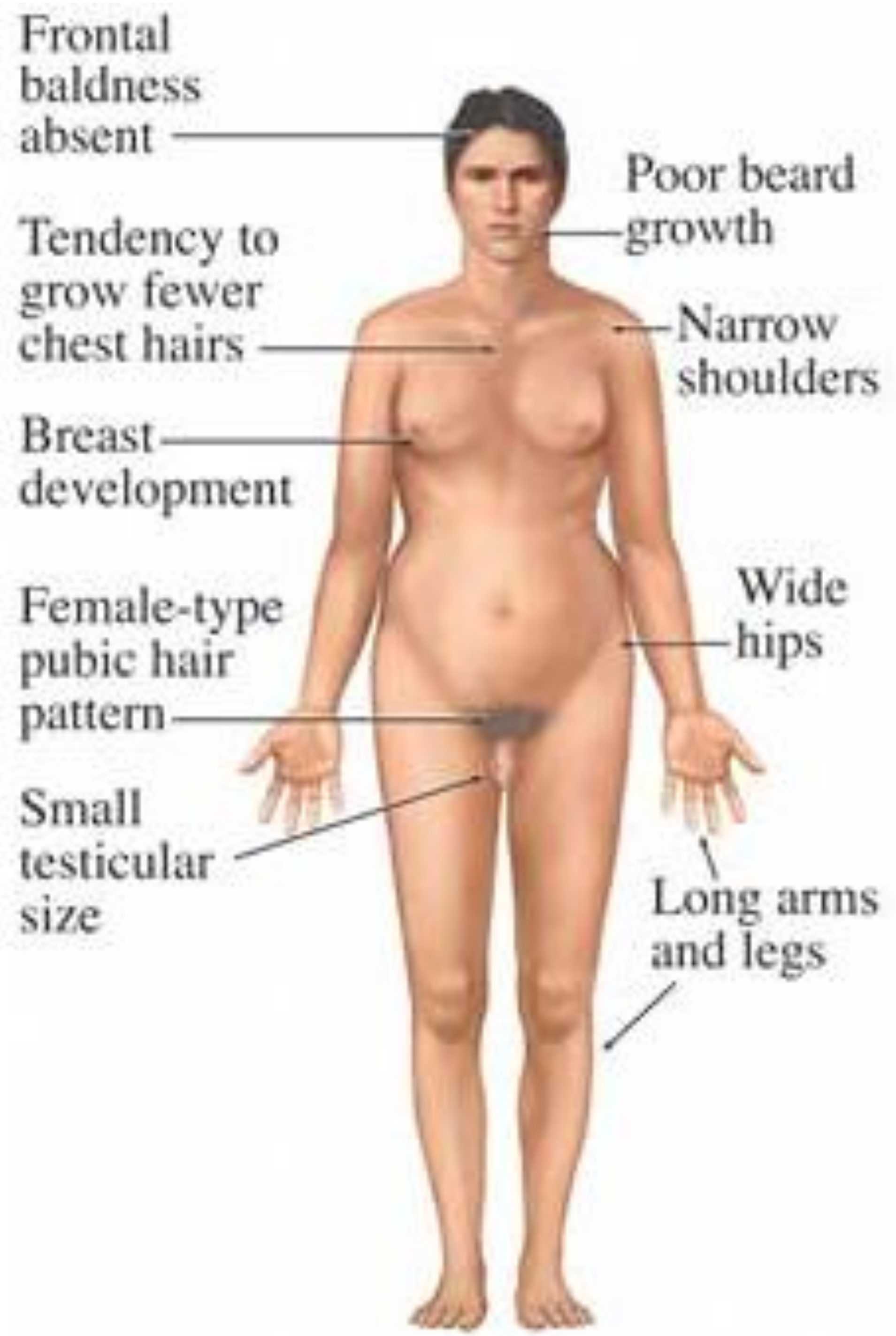
Transposition

- Males with XX karyotype
- Females with XY karyotype

Klinefelter syndrome

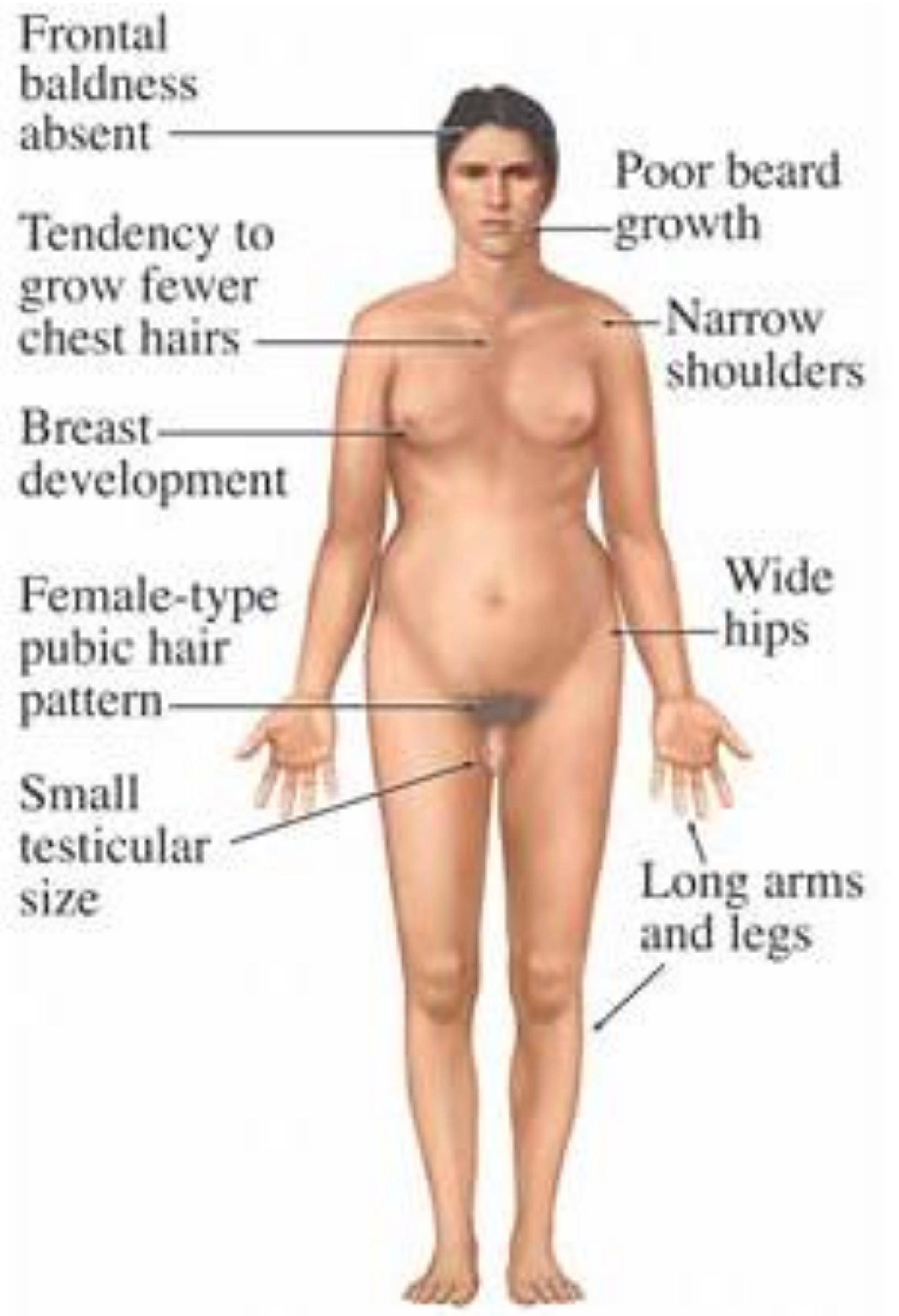
- 47, XXY
- Most common
- Incidence: 1 in 500 males

- Presence of extra X chromosome → phenotypically abnormal male
- Normal male internal and external genitalia
- Poor development of testes with hyalinization of seminiferous tubules → sterility

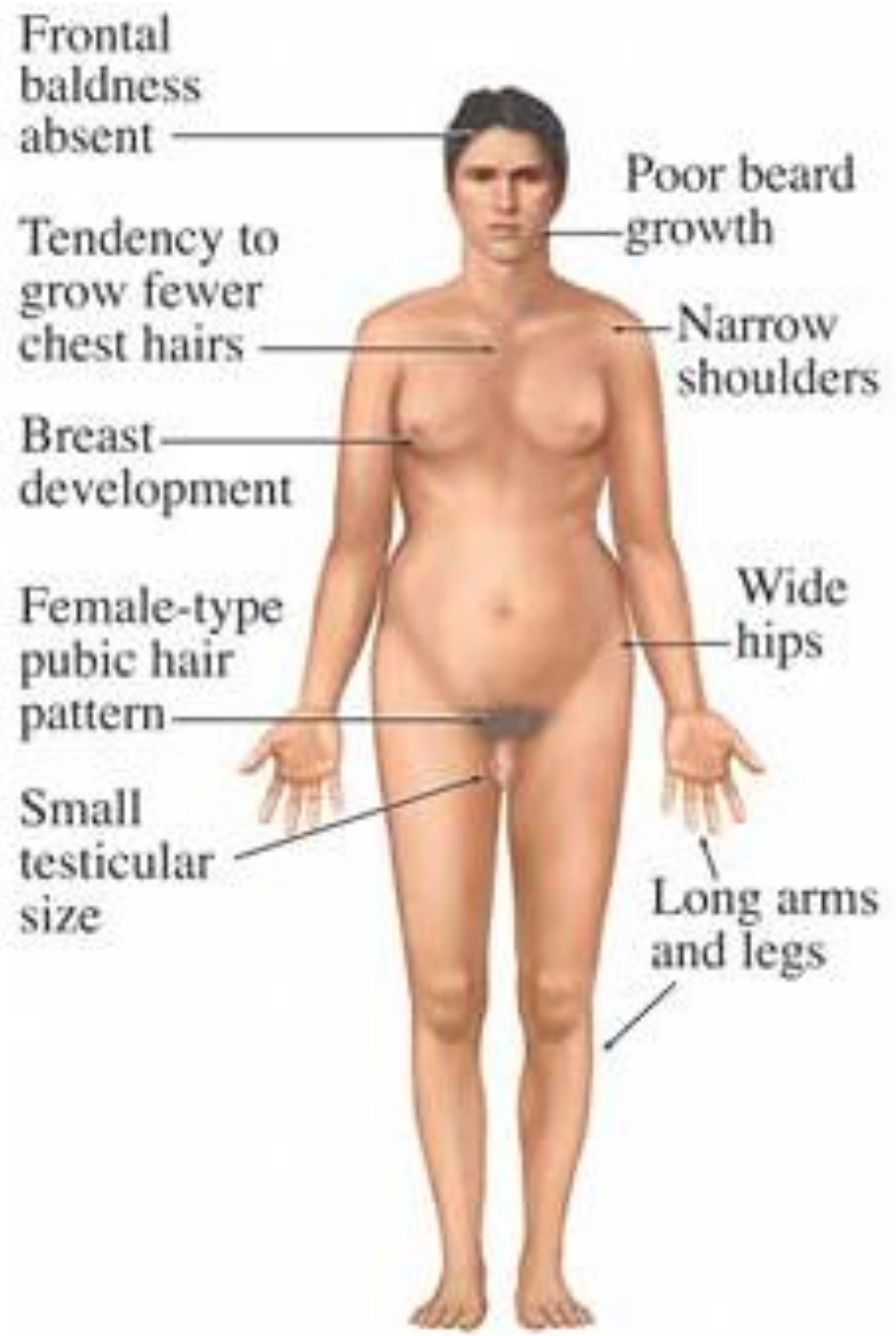


Features:

- Sexual immaturity
  - Primary hypogonadism
  - Testes → very small (pea size)
  - Sterile & impotent
  - Secondary sex characteristics poorly developed → sparse pubic & body hair, small penis

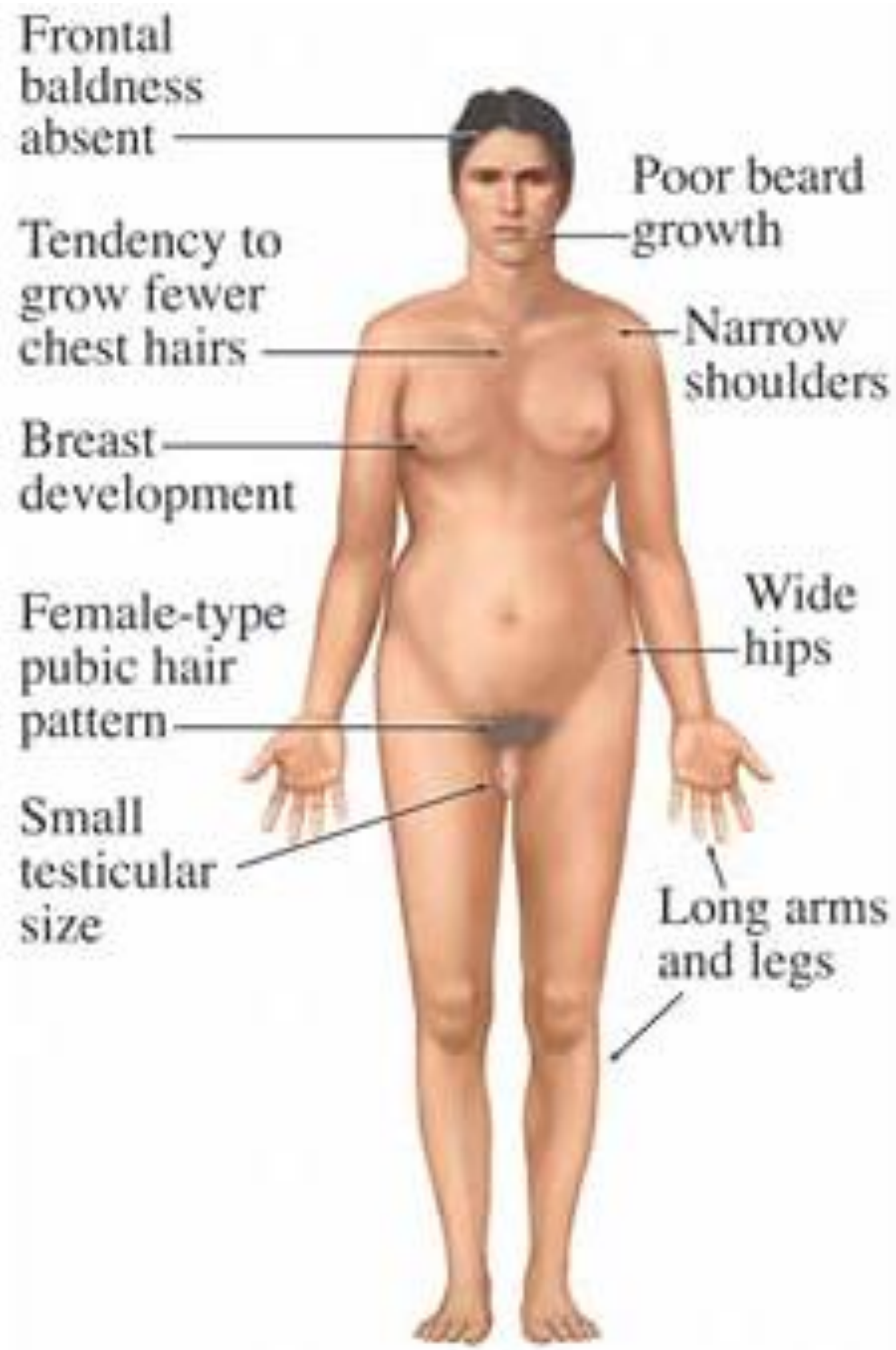


- Narrow shoulders, wide pelvis
- Frontal baldness absent
- Poor beard growth
- Tall → growth of lower segment
- Mental retardation, learning disability +
- Gynaecomastia → development of breast in males
- Increased risk of leukaemias → AML



- Sex chromatin test → positive
- Low or normal plasma testosterone level
- High plasma LH, FSH and Estradiol
- ⇒ Estradiol / testosterone ratio elevated

- Treatment
  - Hormone replacement therapy
  - Fertility treatment
  - Counselling and support for mental health issues



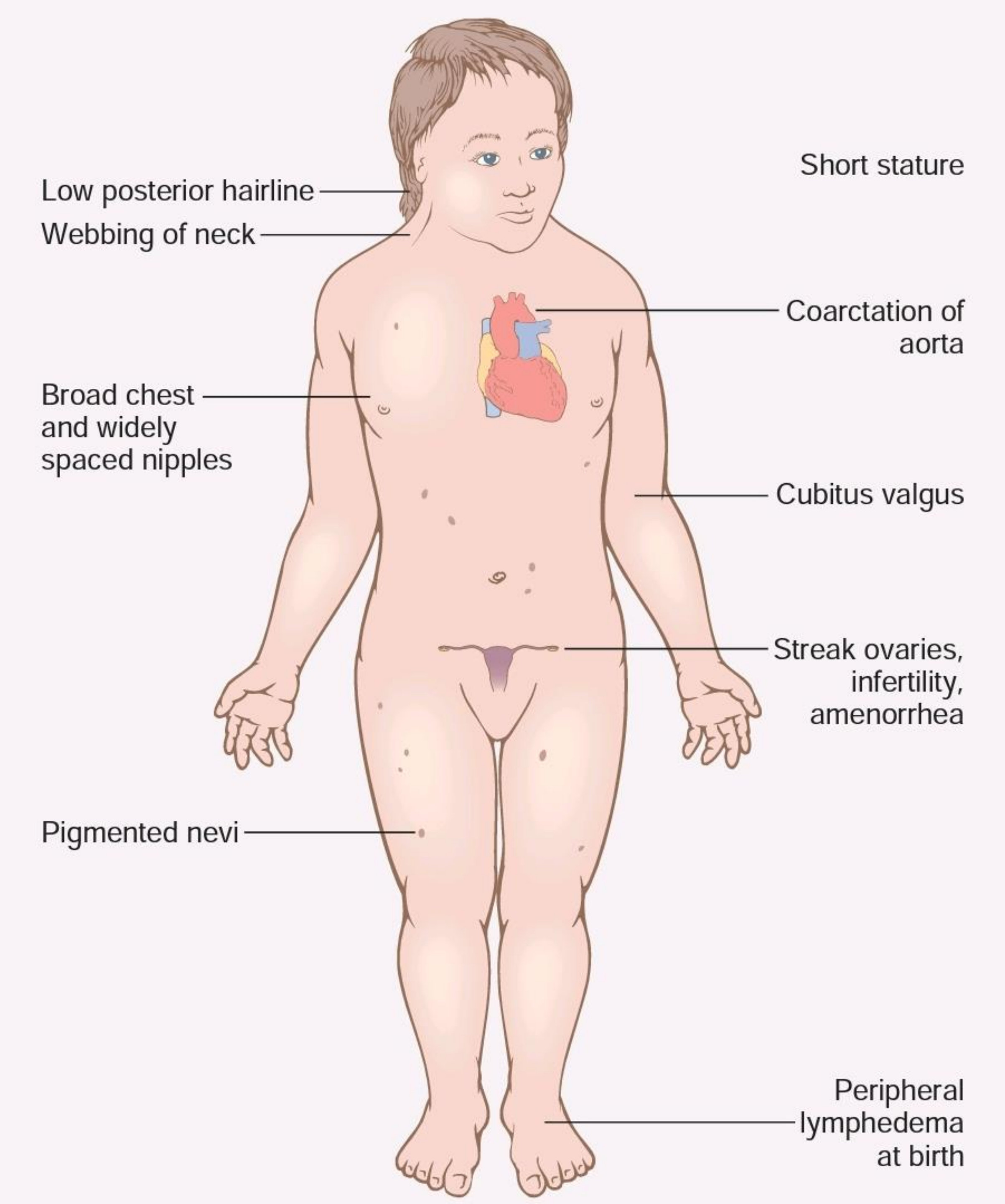
Superfemales

- 47, XXX
- 2nd most frequent sex chromosomal disorder
- Incidence: 1 in 1000 females

- No / mild symptoms → remains undetected
- Poor sexual development
- Scanty menstruation
- Mental retardation

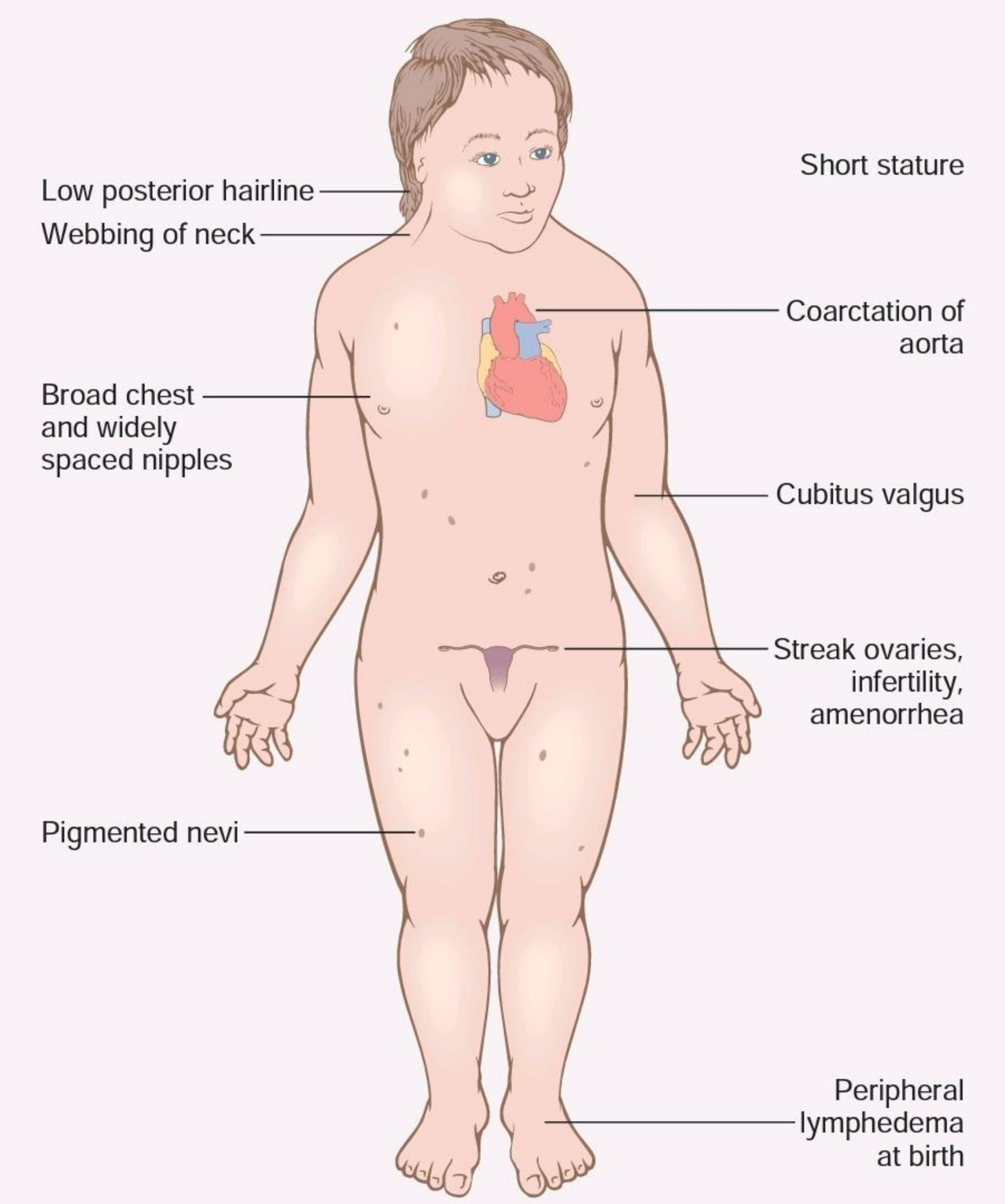
Turner's syndrome

- 45, XO
- Incidence: 1 in 2500 females
- Y chromosome is absent → phenotypically female
- Absent Barr body
- Gonads rudimentary / "streak gonads" due to XO pattern → ovarian dysgenesis

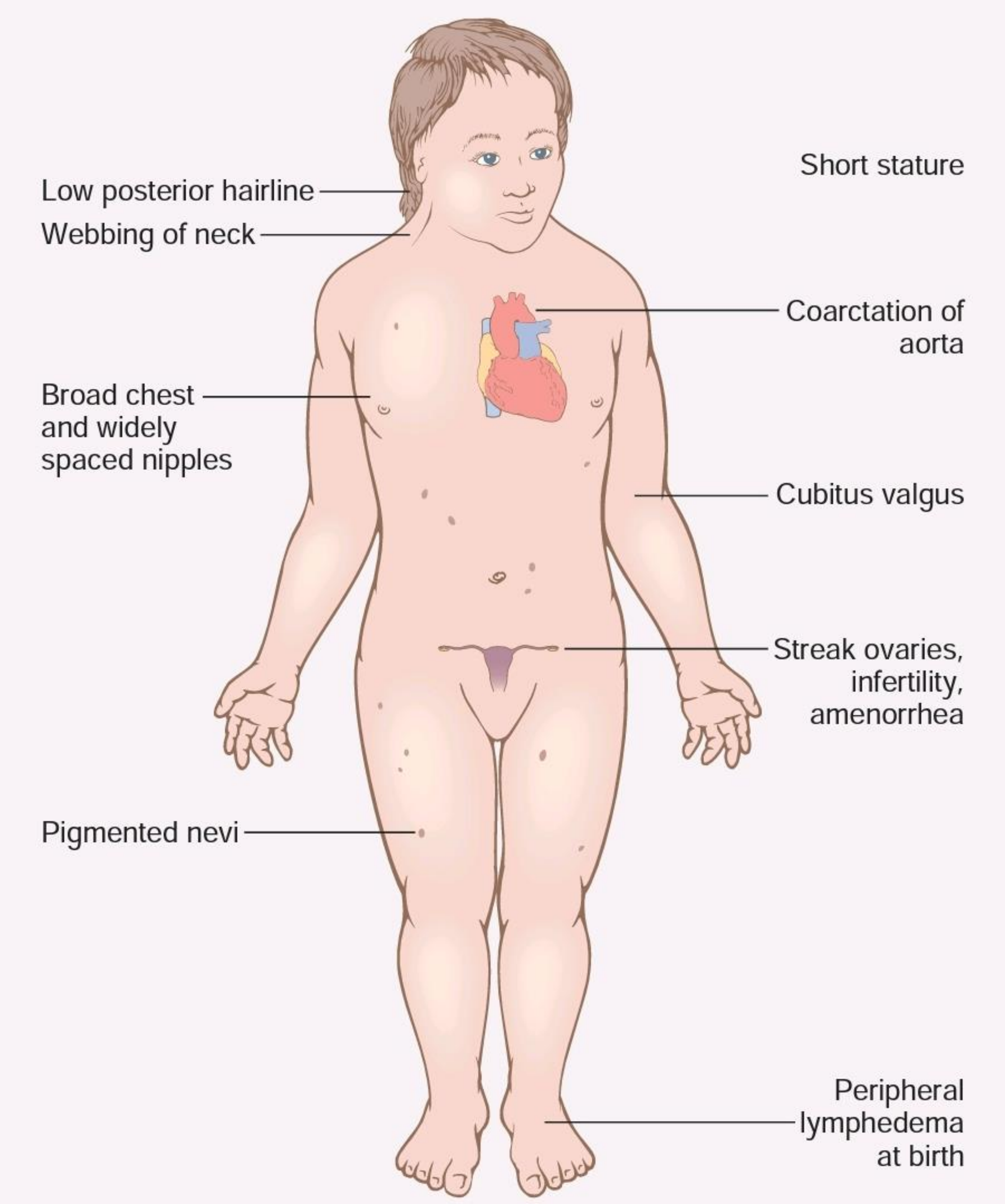


Features:

- Primary infertility
  - Puberty delayed
  - Primary amenorrhoea
  - Scanty menstruation
  - Pubic & axillary hair scanty / absent
  - Vagina & uterus underdeveloped



- Short stature / dwarfism
- Low hairline, low set ears
- Ptosis, exaggerated epicanthic folds
- Webbed short neck, shield like chest, widely separated nipples, amastia
- Low thyroid
- Cubitus valgus, scoliosis
- Lymphoedema of hands and feet
- Short 4th metacarpal

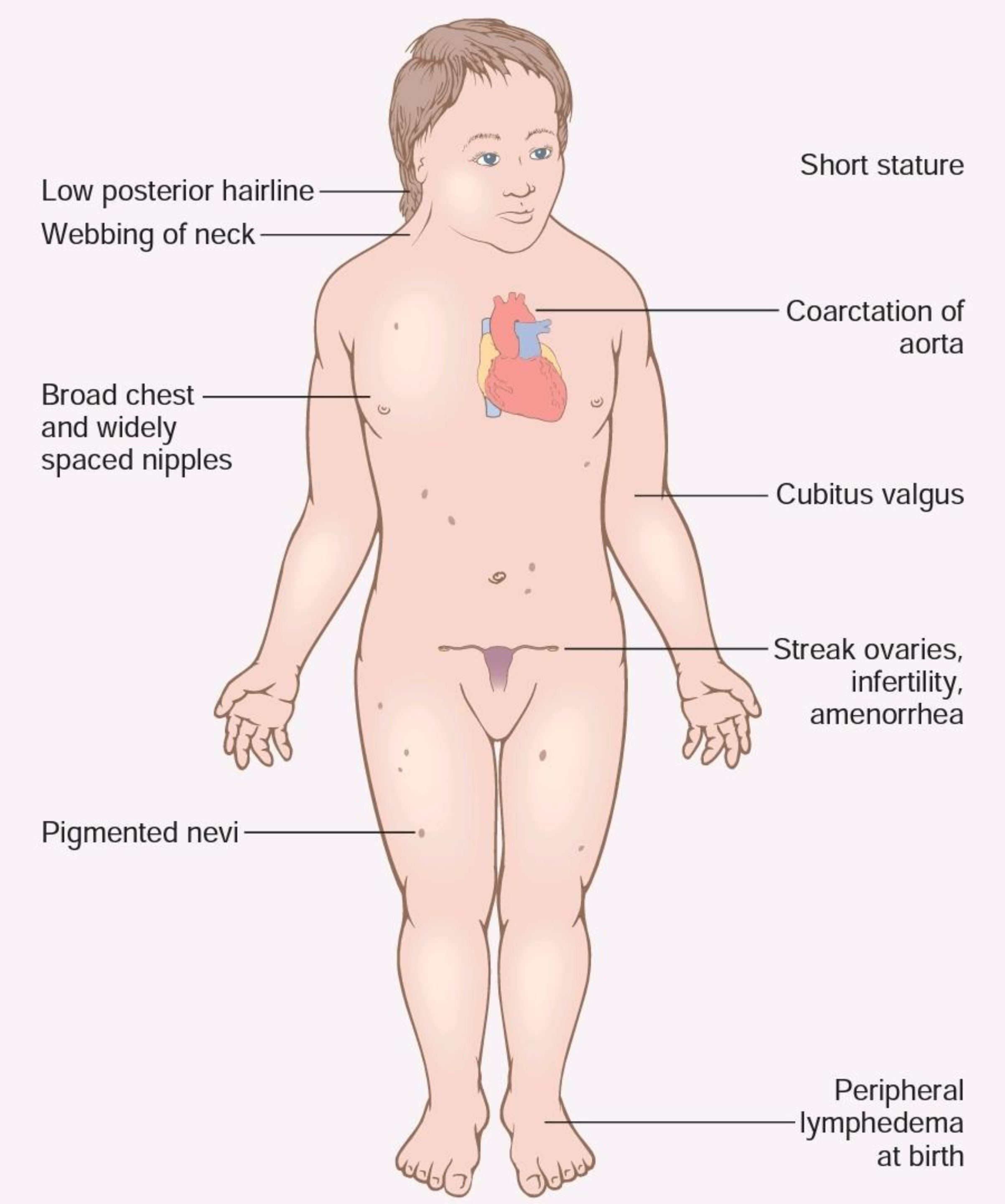


- Cystic fibrosis
- Sensorineural hearing loss
- Coarctation of aorta, high BP

- Treatment
  - Growth hormone therapy
  - Oestrogen therapy

<u>YO combination</u>

- Lethal → intrauterine death of the fetus



Transposition

- Transfer of part of one chromosome to other chromosomes

- ***Males with XX karyotype***
- When short arm of father's Y chromosome gets transposed to father's X chromosome during meiosis → child receives that X chromosome along with mother's X chromosome

- ***Females with XY karyotype***
- Due to deletion of a small portion of Y chromosome containing SRY gene
- So person is XY genetically, but phenotypically female

True hermaphroditism

- Very rare
- Gonads → ovary on one side, testes on other side
- Variations in phenotypic differentiation → combined female & male genitalia

- Hypospadias → external urethral opening on underside of penis or in vagina
- Gonads in labiosacral folds, inguinal canal or abdomen
- Breast development → 60-70% cases
- Chromosomal examination → mixture of 46 XX and 46 XY
- Sex chromatin test → may or may not be positive

Prenatal Diagnosis Of Chromosomal Abnormalities

Amniocentesis

Chorionic Villus Sampling

Developmental abnormalities

Pseudohermaphroditism

- Genetic constitution and gonads of one sex + genitalia of other sex
- 2 forms
  - Male Pseudohermaphroditism
  - Female Pseudohermaphroditism

Female Pseudohermaphroditism

Genotype → XX → female

Gonads → ovaries

Internal genitalia → female

External genitalia → ambiguous genitalia due to virilization

Causes → exposure to androgens around 8th-13th week of gestation

- Congenital virilizing adrenal hyperplasia
- Administration of excess androgens to mother
- Maternal virilizing ovarian tumours (arrhenoblastoma)

Male Pseudohermaphroditism

Genotype → XY → Male

Gonads → Testes

Genitalia → Opposite sex

Causes:

- Defective embryonic testis
- Androgen resistance → 5 α reductase deficiency, mutation of receptor gene

INTERSEX?

TRANSGENDER?

Thank you